Fall Project Proposal #1 – SAFE-Swarm: Search and FInd Emergency Swarm

Names of Team Members, Team Member Role, UCI email addresses:

1. Nader Guirguis, Hardware, nguirgui@uci.edu
2. Samuel Malak, Hardware, swmalak@uci.edu
3. Devin Mccaw, Software, dmccaw@uci.edu
4. Aaron Torres, Software, aptorre2@uci.edu

Updated Project information

Name of Faculty (and other official) advisors:

Boris Vaisband

Weekly check-in mode (in-person, Zoom, email, mixed) and tentative day/time:

In Person, Monday @ 6pm

Updated Summary of Chosen Project

Drone Swarm for Autonomous Search and Rescue. We will build a minimum of 2 drones to demonstrate functionality. The drones will have onboard AI compute for image processing for human detection using a neural network. They will be able to communicate with each other and coordinate. Users (firefighters, search and rescue teams, etc) will be able to specify a search area and drone swarm will search it and flag any humans in the area.

Updated Visual Representation of Project



Updated Major Tasks and Planned WorkLoad Split

- breakdown milestones to individual weekly tasks
- assign each task to 1 responsible member
    - if multiple members needed, break down that task into sub-tasks

- Hardware & Integration (Nader and Samuel)
    - PHASE 1: IKEA-style Drone
        - Making use of off the shelf parts, we'll source, assemble, and flash a drone. We'll use open source software for configuration of drone (PixHawk and Ardupilot)
        - Budget for Drone is ~$250 (part breakdown below)
        - NUCLEO-N657X0-Q Dev board for AI compute, can start working on vision ($75)
    - PHASE 2: Custom Drone
        - Create PCBA for major drone components and integrate AI compute onboard.
        - This would make a more compact, integrated, cheaper board that can be mass produced for swarm expansion
        - Budget for Drone is ~$200

- AI & Swarm Software (Devin and Aaron)
    - (Assumption is that both of us has the knowledge to do any of these things, but roles are assigned in case)

- WINTER DELIVERABLE

    An integrated search-and-rescue demonstration supported by measured performance, documented testing, and complete software documentation.

    - PHASE 1: AI Development and Single-Drone Software

        - Set up the human-detection development environment and prepare labeled training and test images. (Devin)

        - Evaluate a lightweight detection model and measure accuracy and processing speed on a computer. (Devin)

        - Deploy the model to the STM32N6 development board and test camera input and onboard inference. (Devin)

        - Define communication messages for drone identity, position, mission status, and detection reports. (Aaron)

        - Develop firmware for exchanging commands and telemetry between communication modules. (Aaron)

        - Develop a basic interface for users to specify the search-area boundaries. (Aaron)

        - Implement search-area division and waypoint assignment in simulation. (Aaron)

- PHASE 2: Onboard Integration and Swarm Coordination

    - Integrate camera input and human detection with the custom drone hardware, coordinating interfaces with Nader and Samuel. (Devin/Aaron)

    - Package detection results with timestamps and associated drone position for reporting. (Devin/Aaron)

    - Connect mission commands and telemetry to the flight controller, coordinating interfaces with Nader and Samuel. (Aaron)

    - Implement coordinated search assignments for two drones and track mission progress. (Aaron)

    - Implement communication-loss detection and mission-level recovery behavior. (Aaron)

    - Evaluate detection accuracy and onboard inference speed under representative conditions. (Devin)

    - Evaluate communication reliability and search coverage during coordinated testing. (Aaron)

- # Rough Parts List – Autonomous Drone Prototype

Component	Quantity	Estimated Cost
Custom STM32F405 Flight Controller (JLCPCB PCBA)	1	60–100
4-in-1 ESC (35A)	1	$58
Brushless Motors (2004, 2000KV)	4	$54
5–6-inch Propellers	4	$4
GPS + Compass Module (BN-880)	1	$24
ESP32-CAM (OV2640)	1	$7

ESP32 Communication Module	1	$5
4S 3000mAh Li-ion Battery	1	$55
ELRS Radio Receiver	1	$9
Connectors, Capacitors, Wiring	1 set	$3
3D-Printed Frame & Mounts (UCI)	1 set	$0
Estimated Total per Drone		279–**319**

- **Additional Shared Hardware:** ELRS radio transmitter (~$33).
- **Design:** Two Drones, one IKEA-style and one Custom

Estimated Deliverables at Major Milestones (i.e. every ~100 hours of development)

SOFTWARE TIMELINE AND MILESTONE DELIVERABLES:

- Week 3
- - Establish the AI development environment and select an initial human-detection model. (Devin)
- - Establish the software repository and define communication messages and software interfaces. (Aaron)
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- Week 4
- - Prepare training and test images and run baseline human detection on a computer. (Devin)
- - Set up flight simulation and demonstrate a single-drone waypoint mission. (Aaron)
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- Week 5
- - Test camera input on the STM32 development board, subject to component arrival. (Devin)
- - Demonstrate command and telemetry exchange between communication modules, subject to component arrival. (Aaron)
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- Week 6
- - Deploy an initial detection model to the STM32N6 board and record inference performance. (Devin)
- - Implement search-area division and waypoint assignment for two simulated drones. (Aaron)

- - Demonstrate user-entered search boundaries feeding into simulated search-area division and waypoint assignment. (Aaron)
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- Week 7
- - Improve detection performance using representative images and document incorrect or missed detections. (Devin)
- - Implement mission-state tracking, connection monitoring, and communication-loss handling. (Aaron)
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- Week 8
- - Produce detection messages containing confidence scores and timestamps. (Devin)
- - Connect detection messages to mission logging and associate them with available drone-position data. (Aaron)
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- Week 9
- - Test onboard detection with the prototype camera setup and identify integration issues. (Devin)
- - Test communication with the prototype flight controller and demonstrate coordinated search in simulation. (Aaron)
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- Week 10
- - Demonstrate onboard human detection on the development board or prototype and report measured accuracy and processing speed. (Devin)
- - Demonstrate two-agent search simulation, telemetry exchange, and detection-event logging. (Aaron)
- - Complete AI documentation and identify remaining Winter integration tasks. (Devin)
- - Complete communication and mission-software documentation and identify remaining Winter integration tasks. (Aaron)
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- Winter Quarter: Early Development
- - Integrate the AI model and camera pipeline onto the custom drone hardware. (Devin)
- - Integrate mission commands, telemetry, and communication firmware with both drones. (Aaron)
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- Winter Quarter: System Testing
- - Test human detection under representative lighting, viewing angles, and motion. (Devin)
- - Test coordinated search assignments, communication-loss behavior, and mission progress tracking. (Aaron)
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- Winter Quarter: Final Demonstration
- - Validate onboard detection and document final performance results. (Devin)
- - Demonstrate two-drone search coordination and reporting of detected humans, subject to flight readiness. (Aaron)
- - Finalize AI source code, test results, and documentation. (Devin)
- - Finalize communication and swarm source code, test results, and documentation. (Aaron)

Hardware:

- Week 3:
    - Complete Proposal

        - Ordering STM32N6 devboard and camera for beginning AI training
        - Finalize BOM for IKEA-Drone
- Week 5:
    - Have initial version of Schematic and Layout of custom Drone PCBA for review
    - Beginning assembly of IKEA-Drone
- Week 10:
    - Functioning IKEA-Drone (Able to hover)
    - Initial Custom Drone
    - Trained AI model able to recognise humans in realistic test environments
- Winter Quarter
    - Integrated AI stack with Custom Drone
    - Fully tested

Additional Comments / Advisor Feedback

Initial comments:

- Interesting idea, continue planning
- Make rough timeline, initial budget, task breakdown (done)